

Major Assignment-2 (Chemineers)

1. Parameter Computation:-

$$\dot{C}_h = 4.5 \text{ kW/K}; \quad \dot{C}_c = 3.0 \text{ kW/K}; \quad U = 300 \text{ W/(m}^2\text{K)} \\ = 0.3 \text{ kW/(m}^2\text{K)}$$

$$A = 6 \text{ m}^2$$

→ Heat Capacity Rates:-

- $C_{\min} = 3.0 \text{ kW/K}$ (The cold fluid has the smaller rate)
- $C_{\max} = 4.5 \text{ kW/K}$

→ Heat Capacity Ratio (C_r) :-

$$C_r = \frac{C_{\min}}{C_{\max}} = \frac{3.0}{4.5} = 0.667$$

→ Number of Transfer Units (NTU) :-

$$NTU = \frac{UA}{C_{\min}} = \frac{0.3 \times 6}{3.0} = 0.6$$

2. Effectiveness (ϵ) Calculation:-

→ Parallel ~~flow~~ flow heat exchanger:-

$$\epsilon_{\text{parallel}} = \frac{1 - \exp[-NTU(1+C_r)]}{1+C_r}$$

$$= \frac{1 - \exp[-0.6 \times 1.667]}{1.667} \approx 0.379$$

→ Counter flow heat exchanger:-

$$\epsilon_{\text{counter}} = \frac{1 - \exp[-NTU(1-C_r)]}{1 - C_r \exp[-NTU(1-C_r)]}$$

$$= \frac{1 - \exp[-0.6 \times 0.333]}{1 - 0.667 \exp[-0.6 \times 0.333]}$$

$$= 0.399$$

3. Heat Transfer Rate & Outlet Temperatures:-

Max possible heat transfer:

$$Q_{\max} = C_{\min} (T_{h,\text{in}} - T_{c,\text{in}}) = 3.0 \times (120 - 30) = 270 \text{ kW}$$

	$Q = \epsilon Q_{\max}$	$T_{h,\text{out}} = T_{h,\text{in}} - Q/\dot{C}_h$	$T_{c,\text{out}} = T_{c,\text{in}} + Q/\dot{C}_c$
Parallel	102.3 kW	97.3°C	64.1°C
Counter	107.7 kW	96.1°C	65.9°C

5. Why Counter-flow is more effective:- In counter-flow, the temperature difference between the two fluids remains more uniform across the entire length of the exchanger. This maintains a higher average driving force (ΔT). In parallel flow, the two fluids approach the same temperature quickly, causing the driving force to drop off rapidly towards the exit.

Upper limit of Effectiveness:- As NTU increases (usually by increasing area), effectiveness approaches a limit.

- for counter-flow $\epsilon \rightarrow 100\%$, if $C_r < 1$ (the cold fluid could theoretically reach the hot inlet temperature)
- for parallel-flow, $\epsilon \rightarrow \frac{1}{1+C_r}$, in this case parallel flow is physically limited to an effectiveness of roughly 60%, no matter how large you make the heat exchanger.

